Cloud-First Autopilot Zero-Touch Provisioning (Entra Join, Optional Later Domain Join)

Implementation Solution Guide

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Executive Summary — Cloud-First Autopilot Zero-Touch Provisioning (Microsoft Intune)

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Provisioning a new Windows PC the old way is slow and hands-on: a technician images the machine, domain-joins it, installs apps, applies policy, and babysits every build. The modern path is cloud-first — Microsoft Entra join plus Windows Autopilot — so a device provisions itself the moment it is powered on and connected to the internet. Apps and policy arrive during enrollment, the Enrollment Status Page holds the desktop until the device is ready, and the technician does almost nothing. Domain join is added afterward only when a specific app or resource genuinely requires it — never by default.

This is a complete, deploy-it-today package for standing up cloud-first, near zero-touch provisioning with Microsoft Intune, written for an administrator with no prior Autopilot experience.

What you get

- A plain-language Solution Overview of the cloud-first model: Entra join, Autopilot, the Enrollment Status Page, provisioning-time apps and policy, and why this beats hybrid-by-default.
- A Requirements Guide with verify / implement-if-missing for every prerequisite.
- An Implementation Guide of precise, click-by-click Intune steps in the required order — register devices, build the Autopilot profile, configure the ESP and provisioning-time apps, then pilot, validate, and expand.
- A clear decision path for adding domain join later, only when a named app or resource requires it.
- Five PowerShell scripts for readiness discovery, per-device provisioning validation, a provisioning-success and technician-touch report, safe remediation of failed provisioning, and an Autopilot target-group builder.
- Testing and Troubleshooting guides for a confident rollout.

What this delivers

A near zero-touch provisioning model: a dynamic Autopilot device group, a user-driven, Entra-joined deployment profile, an Enrollment Status Page that blocks until apps and policy land, and the core app/baseline set delivered at enrollment — so technicians only unbox and power on. The rare device that truly needs on-premises resources gets domain join added afterward on a documented, evidence-led decision path.

The Phoenixtekk deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

Windows devices that provision themselves on power-on — Entra-joined via Autopilot, apps and policy applied at enrollment, domain join only where truly required — cutting technician effort to almost nothing and proven on a pilot before you scale.

1. Solution Overview — Cloud-First Autopilot Zero-Touch Provisioning

What cloud-first provisioning is

Cloud-first provisioning means a Windows device is set up entirely from the cloud, with no on-premises imaging and no Active Directory domain join. The device is Microsoft Entra joined (not Active Directory domain joined, not Hybrid) and provisioned by Windows Autopilot through Microsoft Intune. The moment a new or reset device is powered on and connected to the internet, it identifies itself to Autopilot, applies your deployment profile, enrolls in Intune, and pulls down apps and policy — while the Enrollment Status Page (ESP) holds the desktop until the device is ready. The result is a near zero-touch experience: the technician unboxes the device, powers it on, and walks away.

Why cloud-first beats hybrid-by-default

Hybrid Azure AD join (domain join plus Entra registration) made sense when apps and resources lived on a domain controller. Today most identity, app, and policy needs are met by Entra and Intune, and hybrid adds real cost: line-of-sight to a domain controller during provisioning, an on-prem Intune Connector, VPN-at-OOBE complexity, and slower, more failure-prone builds. The modern guidance is to go Entra-join-first and add domain join only for the specific device that genuinely needs an on-prem resource. This solution is built around that principle: cloud-first by default, domain join later only if a named app or resource requires it.

How it works

1. Each device is registered to your tenant as an Autopilot device by its hardware hash (uploaded by you, or registered by the OEM/reseller at purchase).

1. A dynamic Autopilot device group automatically gathers all registered devices by their Autopilot ZTDID, so you never hand-add members.

1. A Windows Autopilot deployment profile (user-driven, Microsoft Entra joined) defines the out-of-box experience: branding, account type, and the cloud-only join.

1. An Enrollment Status Page blocks the desktop during provisioning until your required apps and policy install, so users never reach a half-configured machine.

1. Provisioning-time apps and a security baseline are assigned to the Autopilot group so they install during enrollment, not days later.

1. The user signs in once with their work account; the device Entra-joins, Intune-enrolls, and lands fully configured. Technician touch is effectively limited to powering it on.

The near zero-touch experience

- Technician effort: unbox, connect power/network, power on. No imaging, no manual join, no per-app installs.
- User effort: connect to a network and sign in once with their work account.
- Everything else — join, enrollment, apps, policy, baseline — happens automatically behind the ESP.

When (and how) domain join is added later

Domain join is the exception, not the default. Add it only when a specific, named requirement cannot be met cloud-first — for example a legacy line-of-business app that authenticates only via Kerberos to on-prem Active Directory, or a file share / print resource that cannot be reached with Microsoft Entra Kerberos or Cloud Kerberos Trust. The decision criteria and the supported path (Microsoft Entra hybrid join, or Microsoft Entra Kerberos for resource access without changing the join type) are detailed in the Implementation Guide. Default everything to Entra join; carve out the rare exception deliberately and document why.

What you can control

- The join type and out-of-box experience via the Autopilot deployment profile.
- Which apps and policy block provisioning via the Enrollment Status Page.
- The provisioning-time app and baseline set assigned to the Autopilot group.
- Device registration (manual hash upload or OEM/reseller registration).
- A documented exception path to domain join for the rare on-prem dependency.

Benefits

- Near zero technician touch — power on and walk away; no imaging or manual join.
- Faster, more consistent builds — every device provisions the same assigned set.
- Cloud-first identity — Entra join with no domain-controller line-of-sight at OOBE.
- Lower complexity and risk — no on-prem Intune Connector or VPN-at-OOBE unless an exception requires it.
- Provable readiness — the ESP guarantees apps and policy are present before the user reaches the desktop.
- A measurable touch report — see provisioning success rate and where any manual touch crept in.

Limitations (be honest with buyers)

- Autopilot needs each device registered (hardware hash); at scale, OEM/reseller registration is strongly preferred over manual upload.
- The device must reach the internet at the out-of-box experience to provision.

- Pure cloud (Entra-join) does not natively authenticate to on-prem AD resources; those rare cases use Microsoft Entra Kerberos / Cloud Kerberos Trust, or the deliberate domain-join exception.
- Provisioning-time apps that are very large extend the ESP wait; keep the blocking set lean.
- White-glove / pre-provisioning is an advanced add-on and is not the focus of this solution.

The standard Phoenixtekk deployment model

> *Register devices → build the Autopilot profile (Entra join) → configure the ESP and provisioning-time apps → validate the zero-touch flow on a pilot → expand. Phoenixtekk pattern: Assess → Configure → Pilot → Validate → Expand.*

2. Requirements Guide — Cloud-First Autopilot Zero-Touch Provisioning

Verify every requirement before configuring. Each entry: Description · Why · Verify · Implement-if-missing. Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

The named artifacts used throughout this solution (reuse these names verbatim):

- Autopilot group: `Autopilot Devices – All Cloud-First – Provisioning`
- Deployment profile: `Autopilot Profile – Entra Join – Provisioning`
- Enrollment Status Page: `ESP – Cloud-First – Provisioning`
- Provisioning app set: `App Assignments – Core Apps – Provisioning`
- Baseline: `Security Baseline – Cloud-First – Provisioning`

2.1 Licensing

Description: Intune (Plan 1) plus Microsoft Entra ID. Windows Autopilot itself is included. Why: Intune delivers the Autopilot profile, the ESP, and provisioning-time apps/policy; Entra provides the cloud join. Verify: M365 admin center → Billing → Licenses. Implement: Assign M365 E3/E5 or Business Premium (Intune + Entra ID P1 included).

2.2 Automatic enrollment configured

Description: MDM automatic enrollment is on so Entra-joined devices auto-enroll in Intune. Why: Autopilot relies on auto-enrollment to bring the device into Intune during OOBE. Verify: Intune → Devices → Enrollment → Automatic Enrollment; MDM user scope is All (or the pilot group). Implement: Set MDM user scope to All (or the pilot group) and Save.

2.3 Supported Windows OS and edition

Description: Windows 10 (21H2+) or Windows 11, Pro/Enterprise/Education, in out-of-box state. Why: Autopilot, Entra join, and the ESP require a supported build and a non-consumer edition. Verify: `winver`; run `CFAP-Discovery.ps1`. Implement: Reset to OOBE, or upgrade to a supported build/edition before registration.

2.4 Internet access at the out-of-box experience

Description: The device can reach Microsoft cloud endpoints at OOBE (Autopilot, Entra, Intune, Windows Update, app content). Why: Cloud-first provisioning happens entirely over the internet at first boot; no domain-controller line-of-sight is needed. Verify: From a test device at OOBE, confirm internet connectivity; from a managed device reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. Implement: Allow Microsoft's published Autopilot/Intune/Entra URL and IP ranges through proxy/firewall; ensure DHCP/Wi-Fi is available at OOBE.

2.5 Device registration path (hardware hash)

Description: A way to register each device to the tenant as an Autopilot device — OEM/reseller registration at purchase, or manual hardware-hash upload. Why: Autopilot only provisions devices registered to your tenant. Verify: Intune → Devices → Enrollment → Windows → Windows Autopilot → Devices; the device is listed. Run `CFAP-Discovery.ps1` to flag unregistered pilot devices. Implement: For pilots, collect the hash and upload it (see Implementation Step 2); at scale, require OEM/reseller Autopilot registration on the purchase order.

2.6 Provisioning-time app and baseline set

Description: The core apps and a security baseline to be applied during enrollment, assigned to the Autopilot group. Why: A zero-touch device must arrive fully usable; apps/policy must install at provisioning, not later. Verify: Intune → Apps → Windows; confirm the core apps exist and are assigned to `Autopilot Devices – All Cloud-First – Provisioning`. Implement: Add the core apps (Required) and assign the baseline as in Implementation Steps 6–7.

2.7 Pilot device(s)

Description: One or more representative devices to validate the zero-touch flow before fleet rollout. Why: Every Phoenixtekk solution is piloted before expansion. Verify: A reset/new device is available and registered to the Autopilot group. Implement: Reset a spare device to OOBE and register its hash for the pilot.

2.8 (Conditional) On-prem dependency assessment for later domain join

Description: A documented list of any app/resource that genuinely requires on-prem Active Directory. Why: Determines whether the rare domain-join exception is needed; cloud-first is the default. Verify: Review line-of-business apps for Kerberos-to-AD-only auth or on-prem-only file/print needs. Implement: If a true dependency exists, plan the exception path (Microsoft Entra Kerberos / Cloud Kerberos Trust for resource access, or Microsoft Entra hybrid join) per Implementation Step 9.

Checklist

- Intune + Entra ID licensing assigned
- Automatic enrollment (MDM user scope) configured

- Windows 10 21H2+/11 supported edition in OOBE — `CFAP-Discovery.ps1`
- Internet reachable at OOBE
- Device registration path in place (OEM/reseller or manual hash)
- Provisioning-time apps + baseline ready to assign
- Pilot device(s) registered to `Autopilot Devices – All Cloud-First – Provisioning`
- (Conditional) On-prem dependency assessed for any later domain-join exception

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** Phoenixtekk solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every Phoenixtekk solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The Phoenixtekk deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Cloud-First Autopilot Zero-Touch Provisioning

Follow these steps in order. All navigation is in the Microsoft Intune admin center (`https://intune.microsoft.com`) unless stated otherwise.

> *Model: user-driven, Microsoft Entra joined Autopilot, assigned to a dynamic Autopilot device group, with an Enrollment Status Page and provisioning-time apps/policy. Default to cloud-first; add domain join later only for a named on-prem dependency (Step 9).*

Named artifacts (reuse verbatim — these match the Requirements Guide):

- Autopilot group: `Autopilot Devices – All Cloud-First – Provisioning`
- Deployment profile: `Autopilot Profile – Entra Join – Provisioning`
- Enrollment Status Page: `ESP – Cloud-First – Provisioning`
- Provisioning app set: `App Assignments – Core Apps – Provisioning`
- Baseline: `Security Baseline – Cloud-First – Provisioning`

Step 1 — Enable automatic MDM enrollment

1. Go to Devices → Enrollment → Automatic Enrollment.
2. Set MDM user scope to All (or your pilot group). Click Save.

Step 2 — Register the pilot device(s)

1. On each pilot device (in OOBE or already running), run `CFAP-Discovery.ps1` to confirm readiness and

to export the hardware hash CSV, or use the built-in `Get-WindowsAutopilotInfo` from the `WindowsAutopilotIntune` / `Get-WindowsAutopilotInfo` script.

1. Go to Devices → Enrollment → Windows → Windows Autopilot → Devices → Import, then upload the CSV.
2. Click Sync and wait for the device to appear under Windows Autopilot → Devices.

(At scale, have the OEM/reseller register devices at purchase so this step is unnecessary.)

Step 3 — Create the dynamic Autopilot device group

1. Open `https://entra.microsoft.com` → Groups → New group.
2. Group type: Security. Name: `Autopilot Devices – All Cloud-First – Provisioning`.
3. Membership type: Dynamic Device.
4. Add the dynamic membership rule: `(device.devicePhysicalIDs -any (_ -contains "[ZTDId]"))`. Save.
5. Click Create. Registered Autopilot devices join this group automatically.

Step 4 — Configure company branding (optional but recommended)

1. Open `https://entra.microsoft.com` → Company branding → Customize (or Add).
2. Add your logo, background, and sign-in text so the OOBE shows your brand. Save.

Step 5 — Create the Autopilot deployment profile (Entra join)

1. Go to Devices → Enrollment → Windows → Windows Autopilot → Deployment profiles → Create profile →

Windows PC.

1. Name: `Autopilot Profile – Entra Join – Provisioning`. Click Next.
2. On Out-of-box experience (OOBE):

• Deployment mode: User-driven.

• Join to Microsoft Entra ID as: Microsoft Entra joined. (This is the cloud-first choice — not Microsoft Entra hybrid joined.)

• Microsoft Software License Terms: Hide.

• Privacy settings: Hide.

• Hide change account options: Yes.

• User account type: Standard.

• Allow pre-provisioned deployment: No (keep the pilot simple).

• Apply device name template: optionally set a naming template (for example `CF-%SERIAL%`).

1. Click Next → Assignments → Add groups → `Autopilot Devices – All Cloud-First – Provisioning`.
2. Click Next → Create.

Step 6 — Assign the provisioning-time apps

1. Go to Apps → Windows → Add and add each core app the device must arrive with (for example the Microsoft 365 Apps, the company VPN/agent, and any line-of-business app). Name this set, in your notes, `App Assignments – Core Apps – Provisioning`.
1. For each app: Assignments → Required → Add group → `Autopilot Devices – All Cloud-First – Provisioning`. Save.
1. Keep the blocking set lean — large apps extend the Enrollment Status Page wait.

Step 7 — Assign a security baseline

1. Go to Endpoint security → Security baselines → choose the Windows baseline → Create profile.
2. Name: `Security Baseline – Cloud-First – Provisioning`. Accept or tune the defaults. Next.
3. Assignments → Add groups → `Autopilot Devices – All Cloud-First – Provisioning`. Create.

Step 8 — Create the Enrollment Status Page (ESP)

1. Go to Devices → Enrollment → Windows → Enrollment Status Page → Create.
2. Name: `ESP – Cloud-First – Provisioning`. Next.
3. Settings:

• Show app and profile configuration progress: Yes.

• Block device use until all apps and profiles are installed: Yes.

• Allow users to reset device if installation error occurs: Yes.

• Show error when installation takes longer than (minutes): set a sensible limit (for example 60).

• Block device use until these required apps are installed: select only the must-have apps from `App Assignments – Core Apps – Provisioning`.

1. Assignments → Add groups → `Autopilot Devices – All Cloud-First – Provisioning`. Create.

Step 9 — Provision a pilot device (the zero-touch flow)

1. Power on a reset/new registered pilot device and connect it to the internet at OOBE.
2. Your company branding appears; the user signs in once with their work account.
3. The device Microsoft Entra joins, auto-enrolls in Intune, and the ESP runs — installing the provisioning apps and baseline — until the device is ready.
1. The user reaches a fully configured desktop. Technician touch: power-on only.

Step 10 — Validate the pilot

1. Run `CFAP-Validation.ps1` → confirms the device is Autopilot-registered, Microsoft Entra joined,

Intune-enrolled, and that the assigned apps/baseline are present.

1. Run `CFAP-Report.ps1` → confirms provisioning success and the technician-touch summary across pilot

devices.

1. Complete the Testing Guide (Section 5) before expanding.

Step 11 — Decision path: add domain join LATER, only if required

Default to cloud-first. Add domain join only for a device that has a named, verified on-prem dependency. Use these criteria in order:

1. Does any required app authenticate only via Kerberos/NTLM to on-prem Active Directory (no modern

auth, no Entra app)? Does the device need an on-prem-only file share or print server?

- If No → stay Entra joined. Done.
- If Yes → continue.

1. Can the on-prem resource be reached without changing the join type? Prefer Microsoft Entra Kerberos

/ Cloud Kerberos Trust (for SMB/Kerberos resource access from an Entra-joined device). If that meets the need → stay Entra joined and enable Cloud Kerberos Trust. Done.

1. Only if neither cloud option works, adopt the exception: Microsoft Entra hybrid join for that device

class. This requires line-of-sight to a domain controller and the on-prem Intune Connector for Active Directory. Create a separate hybrid Autopilot deployment profile (Join type: Microsoft Entra hybrid joined) and a separate device group; do not change the default cloud-first profile. Document the named dependency that justified the exception.

Step 12 — Expand to production

1. Register production devices (OEM/reseller registration at purchase is ideal at scale).
2. They join `Autopilot Devices – All Cloud-First – Provisioning` automatically (dynamic group) and

inherit the profile, ESP, apps, and baseline.

1. Re-validate each wave with `CFAP-Validation.ps1` and `CFAP-Report.ps1`; track the technician-touch

rate trending toward zero.

Order of operations

`Auto-enrollment → Register devices → Dynamic group → Branding → Deployment profile (Entra join) → Provisioning apps → Baseline → ESP → Provision → Validate → (Domain-join exception only if required) → Expand`

5. Testing Guide — Cloud-First Autopilot Zero-Touch Provisioning

Run after the pilot device provisions (Implementation Step 10) and after each expansion wave.

Test 1 — Device is Autopilot-registered

Do: Intune → Devices → Enrollment → Windows → Windows Autopilot → Devices; confirm the pilot device is listed with the profile assigned. Or run `CFAP-Validation.ps1`. Expected: The device appears and the deployment profile status is Assigned.

Test 2 — Profile applied and join type correct

Do: Run `CFAP-Validation.ps1`, or on the device run `dsregcmd /status`. Expected: `AzureAdJoined: YES` and `DomainJoined: NO` (cloud-first); the assigned profile is `Autopilot Profile – Entra Join – Provisioning`.

Test 3 — ESP blocked and completed

Do: Watch the provisioning run, or check Intune → Devices → Monitor → Enrollment Status Page. Expected: The Enrollment Status Page blocked the desktop, installed the required apps/baseline, and finished with success.

Test 4 — Provisioning-time apps present

Do: Run `CFAP-Validation.ps1` (checks managed-app install state), or open Company Portal on the device. Expected: Every app in `App Assignments – Core Apps – Provisioning` shows Installed.

Test 5 — Security baseline applied

Do: Intune → Endpoint security → Security baselines → `Security Baseline – Cloud-First – Provisioning` → check the device status. Expected: The pilot device shows Succeeded for the baseline.

Test 6 — Near zero-touch confirmed

Do: Record the technician actions performed during the build (target: power-on only). Expected: No imaging, no manual domain join, no per-app installs — technician touch is power-on only.

Test 7 — Provisioning + touch report

Do: Run `CFAP-Report.ps1`. Expected: A clear per-device provisioning-success result and a technician-touch summary across the pilot group.

Test 8 — (Conditional) Domain-join exception, if used

Do: Only if Step 11 created a hybrid exception: on that device run `dsregcmd /status`. Expected: For the exception device only, `DomainJoined: YES` and `AzureAdJoined: YES`; all default devices remain cloud-first.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–7 pass; the pilot device provisions cloud-first with apps/baseline present and technician touch is power-on only
Expand pass	Tests 1–5 and 7 pass for each wave before broadening rollout

6. Troubleshooting Guide — Cloud-First Autopilot Zero-Touch Provisioning

Use `CFAP-Discovery.ps1` / `CFAP-Validation.ps1` for device state, `CFAP-Report.ps1` for fleet status, and `CFAP-Remediation.ps1` (default `-WhatIf`) for safe fixes.

A. Device does not start Autopilot at OOBE

Symptom	Cause	Remediation
OOBE shows a normal consumer setup, no branding	Device not registered, or profile not yet assigned	Confirm under Devices → Enrollment → Windows → Windows Autopilot → Devices; Sync; wait for profile status Assigned; re-OOBE
Device never reaches the cloud	No internet at OOBE	Provide Wi-Fi/DHCP and allow the Autopilot/Intune/Entra endpoints; retry
Registered but no profile applies	Not in the dynamic group, or rule wrong	Run `CFAP-Set-AutopilotTargeting.ps1`; verify the `[ZTDId]` membership rule and group assignment

B. Enrollment Status Page errors or hangs

Symptom	Cause	Remediation
ESP stuck on app install	A required app is large or failing	Trim the ESP blocking set; check the app's install status; run `CFAP-Remediation.ps1` to re-trigger install

Symptom	Cause	Remediation
ESP times out	Timeout too short or app dependency missing	Raise the ESP timeout; confirm app dependencies/supersedence; re-provision
ESP shows error, device unusable	Required app/profile failed	Allow reset on error in the ESP; fix the failing app; reset and re-provision

C. Wrong join type

Symptom	Cause	Remediation
Device is hybrid/domain joined unexpectedly	A hybrid profile or GPO matched the device	Confirm the device is in the cloud-first group only; ensure no hybrid profile/Connector targets it; reset and re-provision with `Autopilot Profile – Entra Join – Provisioning`
`dsregcmd /status` shows `DomainJoined: YES` for a default device	Exception profile mis-scoped	Re-scope the hybrid exception to its own group; keep the default profile Entra-join only

D. Apps or baseline missing after provisioning

• Confirm the apps and `Security Baseline – Cloud-First – Provisioning` are assigned to `Autopilot Devices – All Cloud-First – Provisioning` (device-targeted, Required).

• Sync the device (Settings → Accounts → Access work or school → Info → Sync) or run `CFAP-Remediation.ps1` to force a sync and re-evaluation.

E. On-prem resource unreachable from an Entra-joined device

• This is expected for pure cloud join. Prefer Microsoft Entra Kerberos / Cloud Kerberos Trust for SMB/Kerberos resource access before considering domain join.

• Only if a named dependency cannot be met that way, follow the documented exception path (Implementation Step 11) using a separate hybrid profile and group.

F. Diagnostics quick reference

```
dsregcmd /status                # join state: AzureAdJoined / DomainJoined / MDM enrollment
Get-ComputerInfo | Select-Object WindowsProductName, OsBuildNumber, WindowsEditionId
# Autopilot diagnostics during/after OOBE:
mdmdiagnosticstool.exe -area Autopilot -cab C:\Temp\AutopilotDiag.cab
Get-ItemProperty 'HKLM:\SOFTWARE\Microsoft\Provisioning\Diagnostics\Autopilot' -EA SilentlyContinue
```

During OOBE you can press Shift+F10 for a command prompt to run `dsregcmd /status`.

G. When to escalate

If the device is registered, in the dynamic group, the profile shows Assigned, internet is available at OOBE, and provisioning still fails after a re-sync and re-provision, collect the Autopilot diagnostics cab (`mdmdiagnosticstool.exe -area Autopilot`) and open a Microsoft support case.

7. Best Practices — Cloud-First Autopilot Zero-Touch Provisioning

• Default to Entra join. Make Microsoft Entra joined the standard for every device; treat domain join as a deliberate, documented exception, never the default.

• Register at the source. Push Autopilot registration onto the OEM/reseller at purchase so devices arrive pre-registered — manual hash upload does not scale.

• Use a dynamic Autopilot group. Let `Autopilot Devices – All Cloud-First – Provisioning` gather devices by ZTDID automatically so nothing is hand-added.

• Keep the ESP blocking set lean. Block only the must-have apps; large or non-essential apps belong as Available, not Required, so they don't stall provisioning.

• Assign apps and baseline to the Autopilot group, not to users. Device-targeted assignment guarantees the build is identical regardless of who signs in first.

• Name everything consistently. Reuse the artifact names (profile, ESP, group, app set, baseline) with the `<Purpose> – <Scope> – <Stage>` pattern so audits and handoffs are unambiguous.

• Pilot before you scale. Validate the full zero-touch flow on representative hardware and confirm the pass/fail gate before expanding.

• Measure technician touch. Track the touch rate with `CFAP-Report.ps1`; a number trending to zero is the whole point — chase any device that needed manual intervention.

• Prefer cloud paths for on-prem needs. Before adding domain join, try Microsoft Entra Kerberos / Cloud Kerberos Trust for SMB/Kerberos resource access from Entra-joined devices.

• Isolate any hybrid exception. If you must add domain join, use a separate profile and group; never modify the default cloud-first profile, and record the named dependency that justified it.

• Ensure OOBE internet. Confirm Wi-Fi/DHCP and the required Microsoft endpoints are reachable at first boot — cloud-first provisioning depends on it.

• Document the build. Record the artifact names, the app/baseline set, and any exception decisions for

repeatability and audits.

8. Deliverables Summary — Cloud-First Autopilot Zero-Touch Provisioning

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	The cloud-first model (Entra join + Autopilot), the ESP, provisioning-time apps, the zero-touch experience, and the later-domain-join decision criteria
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered Intune steps: register, profile, ESP, apps/baseline, validate, expand, plus the domain-join exception path
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`CFAP-Discovery.ps1`	Reports Autopilot/enrollment readiness — OS build/edition, OOBE/registration state, hardware-hash export, and tenant registration status (read-only)
`CFAP-Validation.ps1`	Per-device provisioning/enrollment validation — Autopilot registration, Entra join, Intune enrollment, assigned apps/baseline present (read-only)
`CFAP-Report.ps1`	Provisioning-success and technician-touch report across the Autopilot group (Microsoft Graph, read-only)
`CFAP-Remediation.ps1`	Safe remediation of failed provisioning; default `-WhatIf` (re-sync, re-trigger enrollment/app install)
`CFAP-Set-AutopilotTargeting.ps1`	Builds/maintains the dynamic Autopilot target group and confirms profile/ESP assignment (Microsoft Graph)

What the buyer receives

- This implementation guide as a branded Word document and PDF.
- All five PowerShell scripts in a single ZIP.
- A repeatable process following the Phoenixtekk model: Assess → Configure → Pilot → Validate → Expand.

Outcome

Windows devices that provision themselves on power-on — Entra-joined via Autopilot with apps and policy applied at enrollment, domain join only where truly required — cutting technician effort to almost nothing, proven on a pilot before you scale.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

CFAP-Discovery.ps1

Purpose: Read-only. Reports whether a device (and the tenant) is ready for cloud-first Autopilot provisioning: supported OS build/edition, join/enrollment state, hardware-hash capture, and whether the device is already registered as an Autopilot device in the tenant. Optionally exports the hardware hash CSV for manual registration.

How to run: Local readiness on the device being provisioned (no modules required for the local checks): powershell -ExecutionPolicy Bypass -File .\CFAP-Discovery.ps1

CFAP-Remediation.ps1

Purpose: Attempts safe, reversible fixes for a device that did not finish provisioning: force an MDM sync, re-trigger the management/enrollment scheduled tasks so apps and policy re-evaluate, and (via Microsoft Graph, optional) issue a device sync action to the Intune managed device. Runs in -WhatIf mode BY DEFAULT — it prints what it would do and changes nothing until you pass -Execute.

How to run: Preview only (default, no changes): powershell -ExecutionPolicy Bypass -File .\CFAP-Remediation.ps1

CFAP-Report.ps1

Purpose: Read-only. Produces a fleet/group-level report of cloud-first provisioning health and the technician-touch picture: which Autopilot devices have provisioned, their enrollment/join state, profile-assignment status, and an inferred technician-touch classification (zero-touch when the device enrolled via Autopilot and is Entra-joined, versus higher-touch when it was domain-joined or has no Autopilot identity). Uses Microsoft Graph windowsAutopilotDeviceIdentities and managedDevices.

How to run: Requires the Microsoft Graph PowerShell SDK and scopes DeviceManagementServiceConfig.Read.All + DeviceManagementManagedDevices.Read.All + Group.Read.All.

CFAP-Set-AutopilotTargeting.ps1

Purpose: Creates (if missing) the dynamic Microsoft Entra security group that gathers all Autopilot devices by ZTDID, and reports whether the Autopilot deployment profile and the Enrollment Status Page are assigned to that group. With -WhatIf (default), it only previews changes; pass -Execute to create the group / set the membership rule. Profile and ESP assignment are reported; the script does not silently re-point production assignments.

How to run: Requires the Microsoft Graph PowerShell SDK and scopes Group.ReadWrite.All + DeviceManagementServiceConfig.Read.All.

CFAP-Validation.ps1

Purpose: Read-only. Confirms a single device provisioned correctly cloud-first: it is Autopilot-registered, Microsoft Entra joined (and NOT domain joined, unless the documented exception applies), Intune-enrolled, and the assigned provisioning apps / security baseline are present. Combines a local

on-device check with an optional Microsoft Graph lookup of the managed device.

How to run: On the provisioned device (local checks only): powershell -ExecutionPolicy Bypass -File .\CFAP-Validation.ps1